2 Digital literacy: Computers and learning

You will learn

- how computers are used to help people learn
- how computers are used to make discoveries
- how to use online research to help complete a project.



You can use computers to help you in any subject you study. The skills you learn in your computing classes do not just help you to learn while you are at school. They will be important throughout your life. When you leave school, computers will be important in your career.

Computers have been used to make important discoveries. You will learn about some of these discoveries in space exploration, medicine, and in understanding natural disasters such as earthquakes. Perhaps in the future you will contribute to a major discovery.

Later in the unit, you will work in small groups to create a short e-learning course. You will use online research to help you complete the project.



Unplugged

How do you use computers to help you do your schoolwork? Make a list. What software and hardware do you use?

Give examples of times you have used a computer to make work you are proud of.

Learning outcomes: Carry out an online research project; Explain how computers can help with learning and discovery

Did you know?

Scientists in a group of British universities created a 'robot scientist'. They claim it is the first computer to have discovered new scientific knowledge. The robot is called Adam. Adam uses **artificial intelligence** to carry out research. Artificial intelligence is a technique that programs computers to think and work like humans. Adam made a discovery about yeast. The discovery was simple. But it was important because it was made without much human help.





Talk about...

Computers can support your work in many subjects, not just computing. In which subjects do you use computers most?

Are there any subjects where you don't use computers at all? Why do you think that is?

artificial intelligence
assistive technology
computer models
e-learning
guidance computer
interactive video
multimedia
simulation
supercomputer
virtual reality
speech generator

2.1 Using computers for learning

In this lesson

You will learn:

- how computers can help you learn and discover information in all the subjects you study.

Spiral back

Computers are not just for computer studies. You can use computers to help you learn in any subject.

You have already learned some skills that will help you in your studies. You have learned how to:

- find information, to help you make the most of the web
- present information using word-processing, spreadsheet and presentation software.

In Student Book 5 you learned how to use a web browser to find information on the web. The skills you have gained in writing search strings should help you with other subjects like geography and history.

What do you already know?

Here is a reminder of some of the skills you have already learned.

You already know how to:

- **identify key words to use in a search**

  Think about the question you are trying to find answers to. Write your question down and underline the key words. This will help you identify what must be included in your web search.

- **use an age-appropriate web browser**

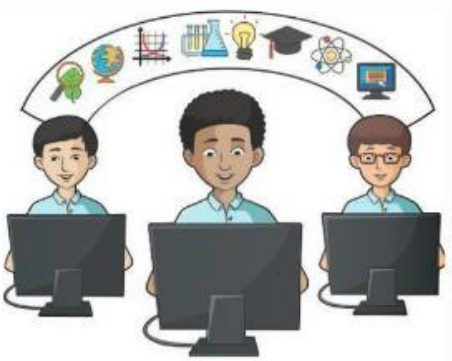
  Using a browser like Kiddle will show you information that is suitable for your age. It will filter out adverts.

- **use special search words and characters**

  You can use special characters in a web browser to make searching easier. For example, you can use the – (minus) sign to exclude terms from the search

- **check the reliability of information you find**

  In Student Book 5, you learned how important it is to check the sources of your information. Can you identify who wrote the information? Does the site publishing the information have a good reputation? Can you check key facts in the information on another site?



- **use bookmarks**

  Bookmarks make sure you can find useful sites again when you need them.

- **give credit for another person's work.**

  If you use content that belongs to another person, you should always indicate where the content comes from.

Now you have started to develop these skills, you can use the web to improve your work in all the subjects you study.

Activity

Choose two subjects that you study other than computing. For each subject, pick a topic that you are studying or have recently studied. Find one or two websites that provide useful information on these topics.

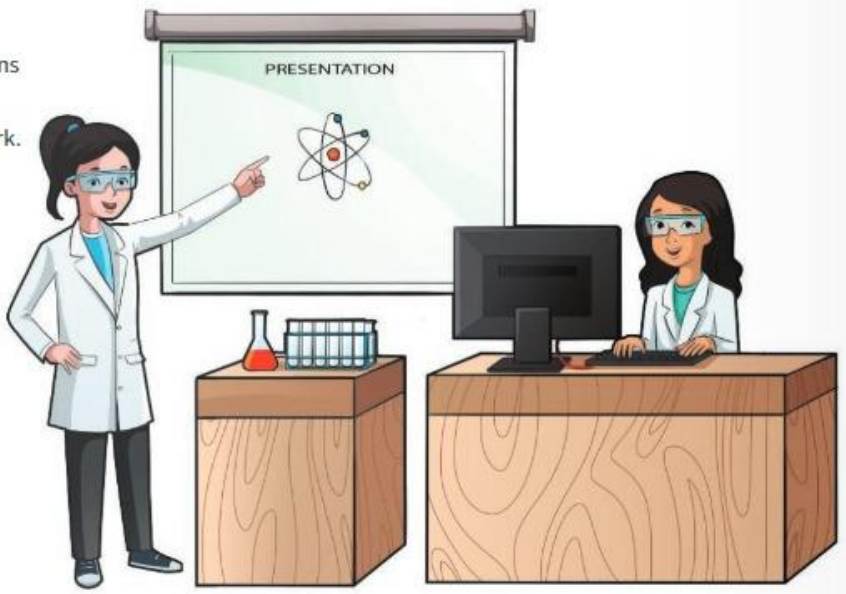
Share the sites with a small group of classmates. Did any of your group suggest other sites you will find useful?

Presenting your work

Think about the skills you have developed using application software. You have used a word processor, a spreadsheet and presentation software. You have also used graphics software to make images. All these skills are useful in any subject you are studying.

You can use application software to:

- write important course notes
- write assignments and homework
- create posters
- make presentations
- add images to illustrate your work.



Explore more

What are the advantages of using a word processor instead of writing your work by hand?

How much do your parents use word-processing apps in their work/at home rather than writing things out by hand? Discuss with them the advantages and disadvantages of using a word processor.

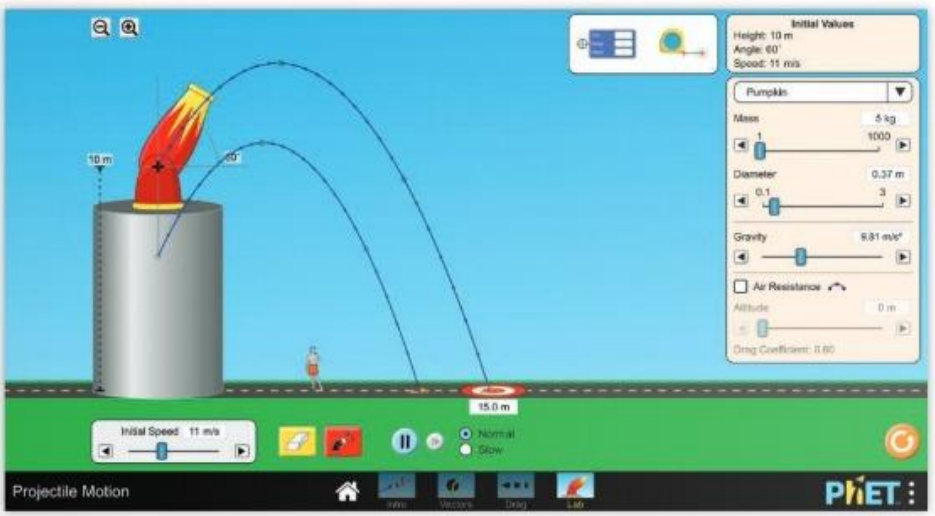
Other uses of technology for learning

Simulations

A computer **simulation** is a program that imitates something you can see or experience in real life. A simulation is like a game you play on your computer. You are in control of objects in a game. You can change how fast they move and the direction they move in.

You can change the way a simulation works. Simulations and games are interactive. This makes simulations a very useful tool for learning.

In science, simulations can be used for experiments. Chemistry experiments with dangerous chemicals can be simulated without students being at risk of injury. In geography, students use simulations to explore how glaciers are formed, or how greenhouse gas emissions affect global warming.



Here are some of the advantages of using simulations.

- They can be done anywhere.
- Experiments can be done many times.
- They speed up processes that take a long time.
- You don't have to set up complicated and expensive equipment.
- They can be used in situations where a real experiment is difficult to do.

Activity

Search the web to find a computer simulation. Use a search string like: 'free computer simulation science elementary school'.

Did you like using the simulation? Why or why not? Was it easy to use?

Virtual reality

Some schools have started to use **virtual reality** (VR) headsets for learning. You may be familiar with virtual reality from computer games. Wearing a VR headset gives you the feeling that you are in a different place. A virtual world is created around you. The world is interactive, like a simulation.

If you are learning about a place in geography or history, you read a description of the place. You can look at pictures or watch videos. With a VR headset, you can step inside the place. You can walk around and explore it.



Assistive technology

Assistive technology helps people with disabilities to learn.

Software can make on-screen text larger and clearer to help students with poor vision. Screen readers read out on-screen text for those with little or no vision. Braille converters translate on-screen text into a pattern of dots called braille. Braille is used as a way of reading by people who have no sight.



Some students have disabilities that make it difficult for them to speak. A **speech generator** turns text that you type into speech, using a simulated voice. The device saves phrases and words that you use often. This saves you from having to re-type everything.



Activity

A mouse can be difficult to use for some people with physical disabilities.

Search the web and find three devices that you can use instead of a mouse.

Extra challenge

Choose one of the devices from the activity.

Make an information sheet for the device you have chosen. Include a picture and write one or two paragraphs to describe how the device works.

Test

1. What is a computer simulation?
2. Give two examples of how assistive technology helps people with disabilities.
3. How can virtual reality be used to help people learn about geography?
4. How do simulations help people learn?

2.2 Using computers for discovery

In this lesson

You will learn:

- how computers can help make discoveries.

Space discovery

The Eagle has landed

On July 20, 1969 the Apollo Lunar Module called Eagle made the first landing on the moon. Neil Armstrong and Buzz Aldrin walked on the moon before returning safely to Earth.



Six Apollo missions to the moon returned safely to Earth. One of the reasons the Apollo missions were safe and successful was the Apollo **Guidance Computer** (AGC).

The AGC was built specially to guide spacecraft to the moon and back. It was the first computer used to guide flights. Today, all aircraft use guidance computers.

Compared with modern computers, the AGC was heavy, slow and difficult to use. It weighed as much as ten modern laptops. The computer chip in a car satellite navigation system is many times more powerful than the AGC.

Activity

There are programs that simulate the Apollo Guidance Computer. They show you how the computer looked and how to enter instructions.

Search the web for 'Apollo Guidance Computer Simulator'. What are the input and output devices it uses? How is it different from the computers you use?

Life on Mars?

Robots are used to explore planets. In January 2004, two identical robot explorers called Spirit and Opportunity landed on Mars. The explorers were supposed to work for 90 days. Opportunity worked for 15 years.

The robot explorers were designed to study the rocks on Mars in the same way that a geologist studies rocks on Earth.



The explorers had:

- two video cameras acting as eyes
- a flexible robot arm to examine rocks
- a hammer to break rocks open
- instruments to analyse rocks
- a microscope to look closely at rocks.

Spirit and Opportunity did not discover life on Mars. They did discover that ancient Mars had water and hot springs that could have supported simple microscopic life.

Activity

Look at the picture of the robot explorer. It seems to have a wing on each side. It doesn't fly.

What are the wings for?

Predicting natural disasters

Weather forecasting

Weather events like hurricanes can be devastating and life-threatening. Understanding how weather events form and move lets scientists called meteorologists predict where bad weather will occur. Early warnings about bad weather gives people time to prepare, or to evacuate an area.

There are two parts to predicting weather.

1. Satellites and radar record real weather conditions. Data about the weather is fed into a computer.
2. The weather data is processed using algorithms that predict how weather conditions will change over the coming hours and days.

This allows meteorologists to predict the course of hurricanes. They can warn people days before the hurricane arrives. The predictions are compared with what really happens. This is so meteorologists can discover more about the way storms work. They change their algorithms to help improve predictions in the future.



Very fast and powerful computers called **supercomputers** are used for this work. A supercomputer can process huge amounts of data very quickly.

Earthquakes

Another type of natural disaster is an earthquake. Parts of the earth move until rocks under the surface break. Huge forces cause the earth to shake violently. Buildings are damaged. Sometimes this causes them to collapse.

Scientists who study earthquakes are called seismologists. Seismologists use sensors to detect movement in the earth. They use satellite images of the earth. They feed this data into a computer model. Algorithms in the model help seismologists discover how earthquakes work.

Scientists now know that an earthquake is going to happen, but they don't know when. They hope that supercomputers will be able to predict earthquakes in the future. Predicting earthquakes will allow people to prepare for them. It will save many lives.



Activity

What preparations could you make if you knew in advance that a hurricane or earthquake was going to happen in your town? How could people be warned?

Medical discovery

DNA

Understanding DNA has led to many important medical discoveries in the 21st century. DNA research involves millions of calculations on large amounts of data. This has only been possible using computers.

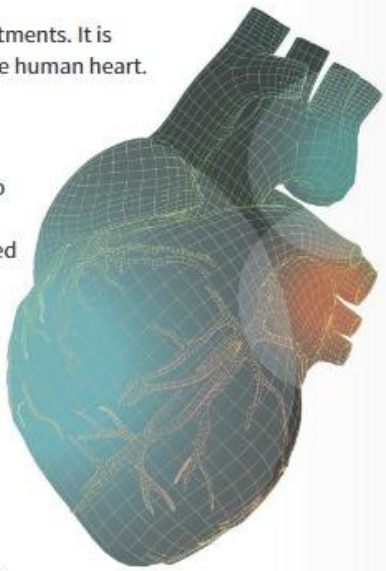
Scientists believe that DNA computers will be developed in the future. A microscopic computer made of DNA will be able to travel through the body and repair cells that are damaged by cancer or other diseases.



Computer models and medicine

Meteorologists and seismologists use **computer models** to make discoveries and to predict events. Other scientists use models too. Doctors and medical scientists use computer models. One use for models is to predict the effect that new treatments will have on patients.

Computer models are used to develop new drugs and other treatments. It is important to know if a new medicine will have a bad effect on the human heart. A computer model can help doctors to find out.



Animal and human testing

Drugs are usually tested on animals before permission is given to use them on humans. Before a new drug can be used, it must be tested on a small group of humans to check it is safe. This is called a clinical trial.

Many people think testing new medicines on animals is cruel. Clinical trials are usually safe but can be dangerous for the people involved. Doctors would like to develop more computer models so that they can test drugs using computers. Some medicines are already tested without using animals.

Activity

Why do you think it is important to use computers to understand and predict natural disasters? Give three reasons.

Extra challenge

Do you think testing medicines on animals is the right thing to do? Would you trust a medicine that has only been tested using a computer?

Outline the advantages and disadvantages of testing medicines using just a computer.

Test

1. Why does a meteorologist use computers?
2. Name one major space discovery that used computers.
3. What are computer models used for in medicine?
4. Name three pieces of equipment used by Mars explorer robots. Can you think of another piece of equipment that would be useful for researching the surface of a planet?